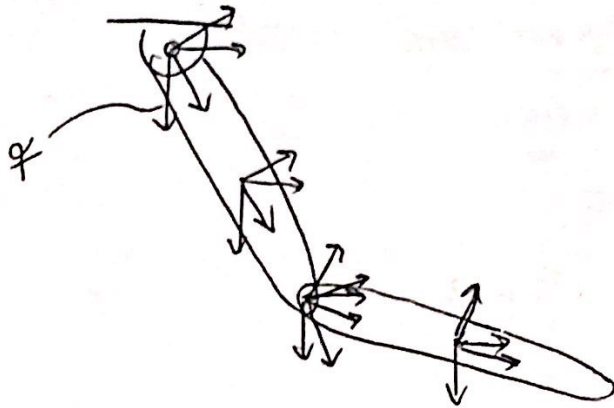


MAE 4730
Lecture 30
11/5/2018

Today:
1) HW Questions
2) Review
3) Rotation

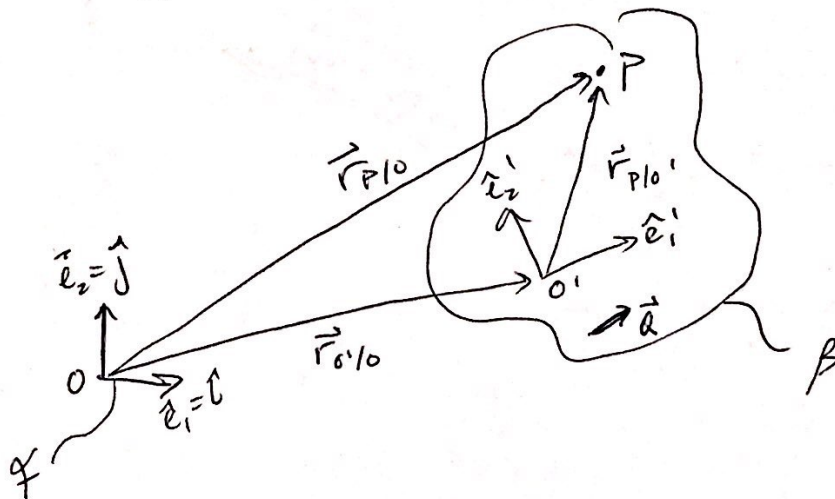


9 reasonable reference frames

Review:

• rot rate stuff

3D

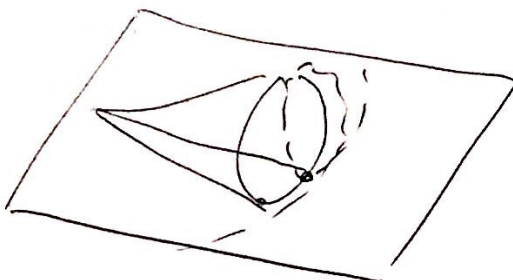


• P can be fixed or not in B

• $\vec{\omega}$ = angular velocity
→ think of as vector that applies to rigid objects
 $= \vec{\omega}_{B/Q}$

Frame derivative: $\boxed{{}^B \dot{\vec{Q}} \equiv \dot{\vec{Q}}' \hat{e}_i'}$ ${}^Q \dot{\vec{Q}} = {}^B \dot{\vec{Q}} + \vec{\omega} \times \vec{Q}$

${}^Q \dot{\vec{\omega}} = {}^B \dot{\vec{\omega}} + \vec{\omega} \times \vec{\omega}$



line of rotation: pt has no velocity
sequence of contact lines

$$\vec{r}_{P/O} = \vec{r}_{O'/O} + \vec{r}_{P/O'}$$

$$\vec{Q} = \vec{Q}$$

$$Q_i \hat{e}_i = Q'_i \hat{e}'_i$$

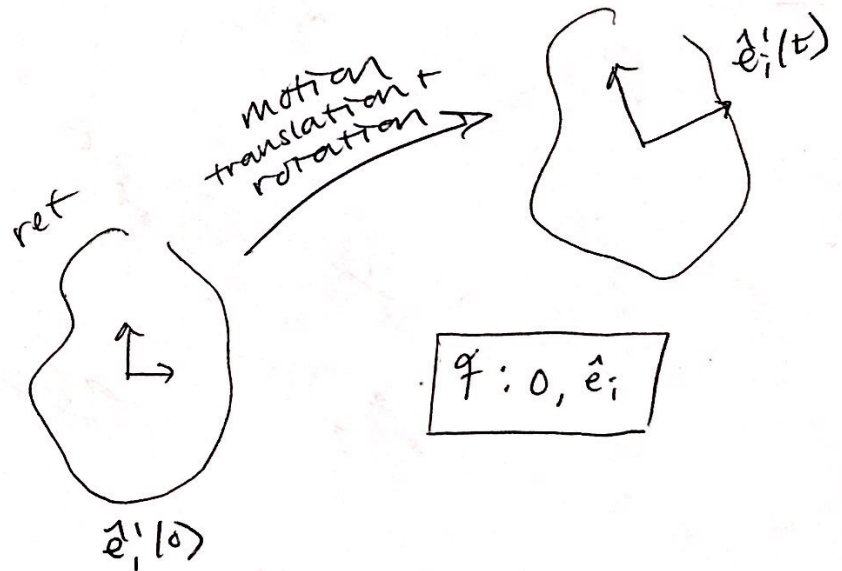
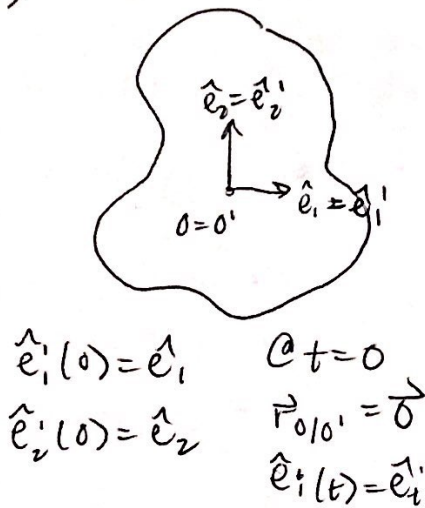
$$\vec{v}_{P/\mathcal{F}} = \vec{v}_{O'/\mathcal{F}} + \vec{\omega} \times \vec{r}_{P/O'} + \vec{v}_{P/B}$$

$$\vec{a}_{P/\mathcal{F}} = \vec{a}_{O'/\mathcal{F}} + \underbrace{\vec{\omega} \times (\vec{\omega} \times \vec{r}_{P/O'})}_{\text{acceleration towards center axis}} + \underbrace{\dot{\vec{\omega}} \times \vec{r}_{P/O'}}_{\text{angular velocity might not be constant}} + \vec{a}_{P/B} + 2\vec{\omega} \times \vec{v}_{P/B}$$

Motion of Rigid Objects

Relative to what?? Anything we're interested in - always use reference configuration coordinates before rotation

3D)



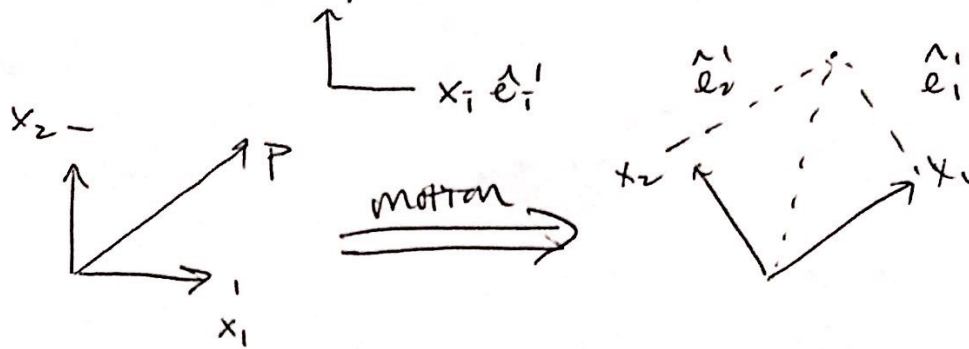
Look at pt. P on β
 ↑
 fixed

@ $t=0$, $\boxed{\vec{r}_P = x_i \hat{e}_i}$

motion ↓

$\vec{r}_P^*$

$$\vec{r}_P^* = \vec{r}_{O'/O} + \vec{r}_{P^*/O'}$$



$\Rightarrow$ motion fully characterized by

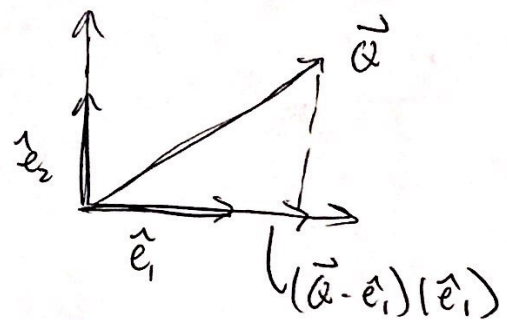
$$\vec{r}_{O'/O} \quad \& \quad \boxed{\begin{matrix} \hat{e}_1', \hat{e}_2', \hat{e}_3' \\ (\hat{e}_1'(t), \hat{e}_2'(t), \hat{e}_3'(t)) \end{matrix}}$$

the rotation

$$\hat{e}_1' = e_{1x}' \hat{e}_1 + e_{1y}' \hat{e}_2 + e_{1z}' \hat{e}_3$$

$$\hat{e}_i' = ((\hat{e}_i') \cdot \hat{e}_j) \hat{e}_j$$

recall $\vec{Q} = (\vec{Q} \cdot \hat{e}_i) (\hat{e}_i)$



Rotation, R

$$R = \begin{bmatrix} \hat{e}_1' & \hat{e}_2' & \hat{e}_3' \\ \text{components} & \text{comps} & \text{comps} \\ \text{of} & \text{of} & \text{of} \\ \hat{e}_1' & \hat{e}_2' & \hat{e}_3' \\ \text{in} & \text{in} & \text{in} \\ \mathcal{F} & \mathcal{F} & \mathcal{F} \end{bmatrix}$$

$$R_{ij} = \hat{e}_i' \cdot \hat{e}_j'$$

$$= \cos \text{ bet. } \hat{e}_i' \text{ \& } \hat{e}_j'$$

$$R = \begin{bmatrix} \cos(\hat{e}_1 \cdot \hat{e}_1') & \dots & \cos(\hat{e}_1 \cdot \hat{e}_3') \\ \vdots & \cos \hat{e}_2 \hat{e}_2' & \vdots \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Rotation matrix:

Q: given x_i such that $\vec{r}_{P/O} = x_i \hat{e}_i$

what are x_i^* such that $\vec{r}_{P/O}^* = x_i^* \hat{e}_i^*$

Know: $\vec{r}_{P/O}^* = x_i \hat{e}_i^*$

$$\begin{aligned}\vec{r}_{P/O}^* &= x_i \hat{e}_i^* \\ &= x_i^* \hat{e}_i\end{aligned}$$

$$\begin{aligned}x_i^* &= \hat{e}_i \cdot \vec{r}_{P/O}^* \\ &= \hat{e}_i \cdot x_j \hat{e}_j^* \\ &= \underbrace{\hat{e}_i \cdot \hat{e}_j^*}_{R_{ij}} x_j\end{aligned}$$

$$\boxed{x_i^* = R_{ij} x_j}$$